

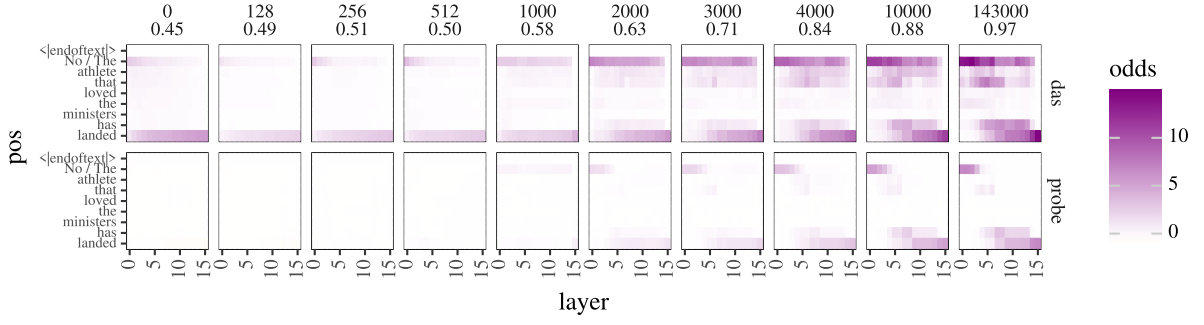


Figure 4: Odds-ratio for checkpoints of pythia-1b on the task `np_i_any_subj_relc`, plotted at every layer and template region. The y -axis is labelled with an example pair of sentences. The plot titles are labelled with the checkpoint and task accuracy. Darker regions indicate a token in a specific layer where causal effect was high.

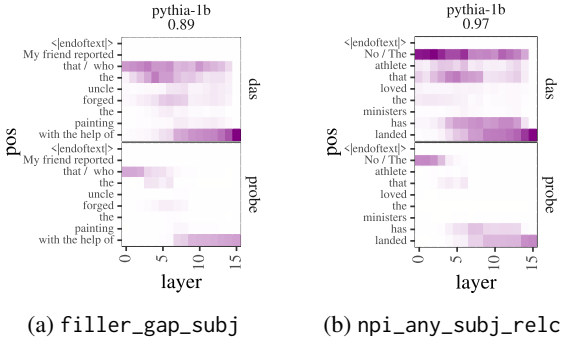


Figure 5: Odds-ratio for each layer and region using DAS and probing on pythia-1b, on two tasks.

replace the label ‘_he’ with ‘_dog’ and ‘_she’ with ‘_give’. We define selectivity for each method by taking the difference between odds-ratios on the original task and the control task for each f , and then compute the overall odds-ratio as in eq. (19).

5.3 Results

We summarise the results for each method in Table 1 by reporting overall odds-ratio and selectivity averaged over all tasks for each model. For a breakdown, see appendices E.1 and E.2.

We find that DAS consistently finds the most causally-efficacious features. The second-best method is probing, followed by difference-in-means. The unsupervised methods PCA and k -means are considerably worse. Despite supervision, LDA barely outperforms random features.

However, DAS is not considerably more selective or (at larger scales) even less selective than probing or diff-in-means; it can perform well on arbitrary input–output mappings. This suggests that its access to the model outputs during training is responsible for much of its advantage.

6 Case studies

In this section, we use **CausalGym** to study how LMs learn negative polarity item (NPI) licensing and wh-extraction from prepositional phrases over the course of training using checkpoints of pythia-1b. We first describe the tasks.

np_i_any_subj_relc. NPIs are lexemes that can only occur in negative-polarity sentential contexts. In this task, we specifically check whether the NPI *any* is correctly licensed by a negated subject, giving minimal pairs like

(20) **No** athlete that loved the ministers has landed $\Rightarrow$ **any**

(21) **The** athlete that loved the ministers has landed $\Rightarrow$ **some**

In (21), where there is no negation at the sentence level, it would be ungrammatical to continue the sentence with the NPI *any*.

filler_gap_subj. Filler–gap dependencies in English occur when interrogatives are extracted out of and placed in front of a clause. The position from which they are extracted must remain empty. The task `filler_gap_subj` requires an LM to apply this rule when extracting from a distant prepositional phrase, e.g.

(22) My friend reported **that** the uncle forged the painting with the help of $\Rightarrow$ **him**

(23) My friend reported **who** the uncle forged the painting with the help of $\Rightarrow$.

In (23), it would be ungrammatical for the preposition to have an explicit object since *who* was extracted from that position, leaving behind a gap.

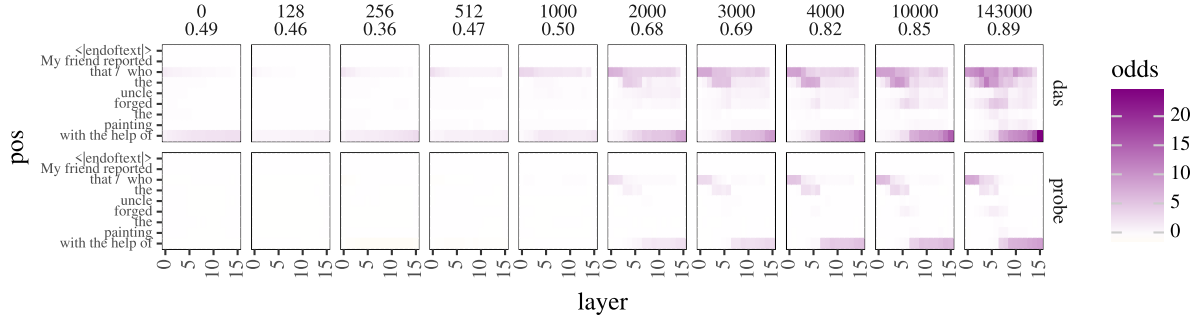


Figure 6: Odds-ratio for checkpoints of pythia-1b on the task `filler_gap_subj`, plotted at every layer and template region.

Final mechanisms. We use the experimental setup of §5.1 and plot the average odds-ratio for each region and layer on the final checkpoint of pythia-1b in Figure 5. For both tasks, we find that the input feature crosses over several different positions before arriving at the output position. For example, in the NPI mechanism (Figure 5b), the negation feature is moved to the complementiser *that* in the early layer, into the auxiliary verb at middle layers, and the main verb in later layers, where its presence is used to predict the NPI *any*. The filler-gap mechanism is similarly complex.

6.1 Training dynamics

To study how the mechanisms emerge over the course of training, we run the exact same experiments on earlier checkpoints of pythia-1b.

npi_any_subj-relc. In Figure 4, the effect first emerges at the NPI (all but last layer) and the main verb (step 1000), then abruptly the auxiliary becomes important at middle layers and the NPI effect is pushed down to early layers (step 2000), and finally another intermediate locations is added at *that* (step 3000). The effect is also distributed across multiple regions in the intermediate layers.

filler_gap_subj. This behaviour takes longer to learn than NPI licensing (Figure 6). The mechanism emerges in two stages: at step 2000, it includes the filler position (*that / who*), the first determiner *the*, and the final token. After step 10K, the main verb is added to the mechanism.

Discussion. For both tasks, the model initially learns to move information directly from the alternating token to the output position. Later in training, intermediate steps are added in the middle layers. DAS finds a greater causal effect across the

board, but both methods largely agree on which regions are the most causally efficacious at each layer. Notably, DAS finds causal effect at all timesteps, even when the model has just been initialised; this corroborates Wu et al.’s (2023) findings.

7 Conclusion

We introduced **CausalGym**, a multi-task benchmark of linguistic behaviours for measuring the causal efficacy of interpretability methods. We showed the impressive performance of distributed alignment search, but also adapted a notion of control tasks to causal evaluation to enable fairer comparison of methods. Finally, we studied how causal effect propagates in training on two linguistic tasks: NPI licensing and filler-gap dependency tracking.

In recent years, much effort has been devoted towards developing causally-grounded methods for understanding neural networks. A probe achieving high classification accuracy provides no guarantee that the model actually distinguishes those classes in downstream computations; evaluating probe directions for causal effect is an intuitive test for whether they reflect features that the model uses downstream. Overall, while methods may come and go, we believe the causal evaluation paradigm will continue to be useful for the field.

A major motivation for releasing **CausalGym** is to encourage computational psycholinguists to move beyond studying the input-output behaviours of LMs. Our case studies in §6 are a basic example of the analysis that new methods permit. Ultimately, understanding how LMs learn linguistic behaviours may offer insights into fundamental properties of language (cf. Kallini et al., 2024; Wilcox et al., 2023b).

We hope that **CausalGym** will encourage compre-

hensive evaluation of new interpretability methods and spur adoption of the interventional paradigm in computational psycholinguistics.

Limitations

While **CausalGym** includes a range of linguistic tasks, there are many non-linguistic behaviours on which we may want to use interpretability methods, and so we encourage future research on a greater variety of tasks. In addition, **CausalGym** includes only English data, and comparable experiments with other languages might yield substantially different results, thereby providing us with a much fuller picture of the causal mechanisms that LMs learn to use. Furthermore, results may differ on other models, since models in the pythia series were trained on the same data in a fixed order; different training data may result in different mechanisms. Finally, justified by the nature of our tasks, we only benchmark methods that operate on one-dimensional linear subspaces; multi-dimensional linear methods as well as non-linear ones await being benchmarked.

Ethics statement

Interpretability is a rapidly-advancing field, and our benchmark results render us optimistic about our ability to someday understand the mechanisms inside complex neural networks. However, successful interpretability methods could be used to justify deployment of language models in high-risk settings (e.g. to autonomously make decisions about human beings) or even manipulate models to produce harmful outputs. Understanding a model does not mean that it is safe to use in every situation, and we caution model deployers and users against uncritical trust in models even if they are found to be interpretable.

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